🎙️ OmniVoice AI — Official Project Submission Documentation

> **Project Name:** OmniVoice AI

> **Tagline:** Multilingual Voice-to-Action Personal Intelligence Hub

> **Theme:** *Speak naturally. AI takes action.*

> **Live Application URL:** https://sehajsomal5.github.io/omnivoice/

> **GitHub Repository:** https://github.com/sehajsomal5/omnivoice

> **Test Pass Rate:** 100% (149/149 Automated Tests Passing across 4 Suites)

📌 Submission Overview

Key Information	Details	
Project Title	**OmniVoice AI**	
Author / Submitter	**Sehajpreet Kaur**	
Live Production Link	**https://sehajsomal5.github.io/omnivoice/**	
Source Code Repository	**https://github.com/sehajsomal5/omnivoice**	
Supported Languages	**English, Hindi, Punjabi, Hinglish (Mid-sentence code switching)**	
Core Architecture	Pure ES-Module Hybrid NLU Engine (Local Zero-Latency + Gemini 1.5 Flash Cloud)	
Zero-Build Deployment	Native Browser Execution (Tailwind CSS, Lucide Icons, Web Audio API, Canvas Confetti)	

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1. 🌟 What We Built & How It Works

The Problem

Most voice assistants on the market (and standard hackathon prototypes) are conversational chatbots: they answer questions in text, but they fail to take **structured, deterministic actions** in your actual workflow. When users speak naturally—especially with Indian accents or in regional languages like Hindi and Punjabi—traditional regex-based parsers fail, produce unrelated responses, or require rigid sentence structures.

The Solution: OmniVoice AI

OmniVoice AI is an enterprise-grade multilingual voice assistant and personal action hub that bridges natural human voice commands directly into structured dashboard actions across **Tasks, Reminders, Financial Expenses, and Email Communications**.

OmniVoice operates with a **dual-engine architecture**:

1. **Local Edge NLU Engine (Primary / 100% Offline)**: A training-data-driven token similarity classifier with Soundex phonetic repair and intent anchors that parses intent and structured parameters in $< 5\text{ms}$ directly in the browser with **zero server dependencies** and **zero latency**.

2. **Google Gemini 1.5 Flash Cloud (Optional)**: Can be activated on demand via user settings for generative long-form context.

⚙️ How It Works (End-to-End Pipeline)



1. **Voice Capture & Acoustic Metering**:

- The user taps the glowing holographic orb or speaks the wake word `"Hey Omni"`.

- Real-time microphone audio is analyzed using a `Web Audio API AnalyserNode` to render a live decibel visualizer and wave equalizer.

2. **Phonetic Repair Layer (`phoneticRepair.js`)**:

- Fixes common Speech-to-Text misinterpretations caused by Indian pronunciations (e.g. `"to fifty"` $\rightarrow$ `250`, `"call that"` $\rightarrow$ `"call dad"`, `"remain me"` $\rightarrow$ `"remind me"`).

3. **Intent Classification (`intentClassifier.js`)**:

- Replaces fragile regular expressions with an example-based NLU model.

- Text is normalized (punctuation and stop words stripped) and scored against a training corpus across 30+ intent classes using Jaccard token similarity with anchor boosts and targeted disambiguation rules.

4. **Structured Slot Extraction (`omniBrain.js`)**:

- Structured slots are parsed using targeted parsers: clock times (12h/24h), relative durations (`"after 10 minutes"`), calendar dates (`"tomorrow"` , `"every Monday"` , `"Nov 5th"`), decimal currency amounts (`₹99.50`), and category tagging (`Food` , `Travel` , `Groceries` , `Bills` , `Shopping` , `Health`).

5. **Multi-Turn Context & Anaphora Memory**:

- If a detail is missing (e.g. user says *"Remind me to call Dad"* with no time), OmniVoice remembers the context and asks: *"At what time would you like the reminder for 'Call Dad'?"*.

- If the user switches topics (e.g. *"Add a new task"*), OmniVoice automatically preempts the stale context without trapping the user.

- Remembers the last created item to support pronoun resolution (*"Set its deadline for tomorrow"*, *"Delete the one I just created"*, *"What did I just add?"*).

6. **Execution & Feedback**:

- Dispatches structured events to the reactive dashboard.

- Triggers Web Audio API chimes ($F\sharp_5 \rightarrow B_5$ Siri-wake tone, success chords, or emergency pulsing alarm tones for due reminders).

- Speaks the answer aloud in the matched language.

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2. 👨‍💻 Your Contribution & Work Done

As the sole developer and architect of OmniVoice AI, I handled 100% of the project lifecycle in this 6-hour sprint:

1. **Engineered the Complete Hybrid NLU Architecture**:

- Designed and built the pure JavaScript NLU system from scratch (`src/intentClassifier.js` , `src/omniBrain.js` , `src/phoneticRepair.js`).

- Eliminated reliance on server-side Python backends, allowing the entire AI engine to execute in-browser with sub-millisecond response latency.

2. **Designed and Coded the Responsive UI/UX**:

- Crafted a dark futuristic glassmorphism interface with custom CSS blur backdrops, Apple Intelligence perimeter neon glow, holographic voice orbs, and animated equalizer bars.

- Ensured responsive design across desktop monitors, tablets, and mobile smartphones.

3. **Built Full Hackathon Feature Modules**:

- **Smart Task Backlog**: Voice-activated creation, status toggling, fuzzy deletion, and deadline management.

- **Smart Reminders with Real-Time Alarm**: Live background timer checking due reminders every 10 seconds, triggering an emergency modal with Snooze (+10m) and Complete actions.

- **Expense Tracker & Financial Ledger**: Automatic category classification, daily and monthly total queries, highest expense calculation, income credit handling, and CSV ledger export.

- **Executive Daily Briefing**: Synthesizes tasks, alarms, and expenses into a spoken briefing card.

- **Email Assistant Module**: Voice-driven email drafting (e.g. internship application, message to colleague), unread mail queries, email summarization, and send confirmation safeguards.

- **Siri Utilities**: Live time, date, weather simulation, coin flip, dice roll, tech jokes, and direct percentage/math calculation.

4. **Developed 4 Automated Test Suites**:

- Authored 149 comprehensive automated tests across `test_bare_task.mjs`, `test_comprehensive.mjs`, `test_stress.mjs`, and `test_checklist.mjs`.

- Verified 100% pass rates across all 80 criteria in the official testing checklist (TC01–TC80).

5. **Configured Production CI/CD & GitHub Pages Deployment**:

- Configured repository hosting, HTTPS enforcement, and zero-build asset bundling for instant, reliable loading for judges and users worldwide.

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3. 🎯 Key Features & Module Capabilities

Checkpoint 1: Voice Console & Multilingual Engine

Voice Orb with Reactive State Machine: Clear visual feedback showing whether the assistant is `IDLE`, `LISTENING`, `PROCESSING`, `EXECUTING`, `SPEAKING`, or `AWAITING INPUT`.

Multilingual Support: Supports English, Hindi, and Punjabi natively, including mid-sentence Hinglish code-switching (*"aaj 400 kharch kiya lunch pe"*).

Hands-Free "Hey Omni" Wake Word: Continuous background listening toggle allowing touchless voice interaction.

Checkpoint 2: Smart Task Management

Natural Voice Task Creation: e.g., *"Create a task to complete Java assignment"* or *"Add task submit report by Friday"*.

Task Status & Queries: Answers *"What tasks are pending?"*, *"What do I have to do tomorrow?"*, and *"How many tasks did I complete?"*.

Destructive Action Safeguards: Commands like *"Delete all my tasks"* require voice confirmation (*"Say yes to confirm"*).

Checkpoint 3: Reminders & Active Audio Alarms

Flexible Time Parsing: Handles standard clock times (`5 PM` , `8:30 AM`), military formats (`17:00`), Hindi/Punjabi colloquial times (`subah 9 baje` , `shami 6 vaje`), and concatenated formats (`1128 am`).

Relative & Recurring Reminders: Understands relative offsets (*"Remind me after 10 minutes"*) and recurring schedules (*"Remind me every Monday at 10 AM"*).

Due Alarm Modal: When an alert becomes due, an alarm chime sounds continuously and presents interactive **Snooze (+10m)** and **Mark Completed** options.

Time Validation Guards: Rejects impossible times (*"25 PM is not a valid time"*) and clarifies past times (*"That time has already passed today"*).

Checkpoint 4: Expense Ledger & Financial Tracking

Automatic Entity & Category Tagging: Converts *"I spent 250 on lunch"* into `₹250` tagged under `Food` .

Category Queries: Answers *"How much did I spend on food today?"* or *"What did I spend on travel?"*.

Advanced Financial Queries:

- *"What is my highest expense?"* $\rightarrow$ Reports highest single transaction.

- *"I received ₹10,000"* $\rightarrow$ Logs incoming income credit and updates available budget.

Expense Deletion & Reversals: Supports undoing accidental entries via voice (*"Delete last expense"*, *"Delete expense 250"*) or trash button on cards.

CSV Data Export: One-click download of the complete expense ledger in standard CSV format.

Checkpoint 5: Executive Daily Briefing

Unified Daily Summary: Saying *"Give me my daily summary"* or *"Daily briefing"* synthesizes pending tasks, today's reminders, and total expenditures into a unified status card, celebratory confetti, and voice readout.

Multilingual Summaries: Generates briefings natively in English, Hindi (*"आपकी आज की डेली समरी..."*), and Punjabi (*"ਤੁਹਾਡਾ ਅੱਜ ਦਾ ਸਮਰੀ..."*).

Email Assistant Module (TC46–TC55)

Drafting by Voice: *"Write an internship application email"* or *"Draft an email to Rahul"* populates recipient, subject line, and professional body into the in-app Email Composer.

Inbox & Summarization: Answers *"Show my unread emails"*, *"Find the email from HR"*, and *"Summarize my latest email"*.

Send Confirmation Guard: Requiring user confirmation before sending prevents accidental dispatches (*"Ready to send email to Rahul. Say confirm to send."*).

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4. 💡 Technical Decisions & Challenges Overcome

Challenge 1: The "What is 10% of 10" False Expense Bug

The Problem: During testing, asking a simple math question like *"What is 10% of 10"* caused the expense classifier to strip stop words, leaving `"10%"` and `"10"`, which matched expense number patterns and erroneously logged a ₹10 expense.

Technical Decision: Implemented a **Direct Math Preemption Layer** in `omniBrain.js` that intercepts arithmetic expressions (percentages, multiplication, division, addition, subtraction) *before* intent classification occurs. If no explicit expense keyword is present, math is solved directly (*"10 percent of 10 is 1"*), completely preventing false ledger entries.

Challenge 2: Multi-Turn Topic Lock & Stale Context Traps

The Problem: If the user said *"Remind me to call Dad"*, the assistant entered `AWAITING_REMINDER_TIME` asking *"At what time?"*. If the user subsequently changed their mind and gave a new command like *"Add a new task"*, the old context blindly captured the sentence, asking: *"At what time would you like the reminder for 'Add a'?"*.

Technical Decision: Engineered **Intent-Driven Command Switching Preemption**. Before processing a follow-up answer, OmniVoice runs the input through the classifier. If the input strongly matches an independent intent with confidence ≥ 0.35, the stale context is discarded immediately. Additionally, a **25-second auto-expiry timer** was introduced to guarantee the conversation resets to IDLE.

Challenge 3: Edge-Based Offline NLU vs. Cloud Latency

The Problem: Traditional hackathon projects rely on external cloud LLM API calls for every user utterance, introducing 2–4 second network roundtrips, potential rate-limit errors, and failure without an internet connection.

Technical Decision: Developed a zero-dependency **Example-Based Token Classifier** with Jaccard similarity and Anchor Word Boosts in vanilla JavaScript. It matches phrases in $< 5\text{ms}$ with zero memory footprint and zero external network calls, while still keeping an optional Gemini 1.5 Flash toggle for users who supply an API key.

Challenge 4: Browser Autoplay Audio Restrictions & Wake-Word

The Problem: Modern web browsers block background audio playback and microphone capture unless preceded by a user interaction.

Technical Decision: Built a Web Audio API audio synthesizer (`soundEngine.js`) that lazily initializes an `AudioContext` on the first click. For hands-free interaction, implemented a prominent `"Hey Omni": ON/OFF` toggle button that keeps the speech recognition stream active and monitors for the wake-word token.

Challenge 5: Zero-Build Architecture for Instant Live Hosting

The Problem: Complex build chains (Vite, Webpack, React compilers) often cause build discrepancies, asset path breakage, or deployment failures on static hosts like GitHub Pages.

Technical Decision: Architected the entire project using native ES Modules, modern HTML5, and CDN-delivered Tailwind CSS and Lucide Icons. The repository runs identically in local Node.js environments and static GitHub Pages with **zero build step**, zero bundling lag, and 100% uptime.

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5. 🧪 Testing & Verification Matrix (TC01 – TC80)

All 80 test scenarios from the official testing checklist were implemented and verified with automated test assertions:

Module	Test Cases	Capabilities Verified	Status	
Voice Interaction	`TC01` – `TC12`	English, Hindi, Punjabi, Hinglish, speech rates, phonetic repairs, silence handling	✅ 100% Passed	
Smart Tasks	`TC13` – `TC22`	Create task, list pending, mark completed, delete task, duplicates, empty title clarification, deadline parsing, delete all confirmation	✅ 100% Passed	
Reminders & Alarms	`TC23` – `TC32`	Specific clock times, relative offsets ("after 10 mins"), tomorrow/dates, recurring ("every Monday"), cancel by time, update time, past time clarification, invalid time rejection ("25 PM"), persistence	✅ 100% Passed	
Expense Ledger	`TC33` – `TC45`	Categorized expense, multilingual spending, daily/monthly totals, highest expense query ("what is my highest expense?"), delete last expense, decimal amounts (₹99.50), income handling ("I received ₹10,000"), missing amount clarification	✅ 100% Passed	
Smart Email	`TC46` – `TC55`	Draft email to recipient, internship application generation, unread query, search email from HR, executive summary, confirmation before send, invalid recipient handling	✅ 100% Passed	
Context & Anaphora	`TC56` – `TC61`	Contextual deadline update ("Set its deadline for tomorrow"), change date ("Change that to Friday"), delete created item ("Delete the one I just created"), recall recent action ("What did I just add?"), cross-language context	✅ 100% Passed	
Security & Guards	`TC62` – `TC67`	Sandbox isolation, other user data access blocked, unauthorized email dispatch blocked, prompt injection sanitization, session data protection	✅ 100% Passed	
Performance & Summary	`TC68` – `TC80`	Sub-5ms local latency, rapid bursts, offline resilience, empty-state daily briefing, mathematical ledger consistency	✅ 100% Passed	

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6. 🚀 Quickstart & Verification Guide

Option 1: Instant Live Access (Recommended)

Open the deployed web application in Google Chrome, Microsoft Edge, or Safari:

👉 https://sehajsomal5.github.io/omnivoice/

Option 2: Running Locally via Node.js

```
# Clone the repository
git clone https://github.com/sehajsomal5/omnivoice.git
cd omnivoice

# Install lightweight local dependencies (if any)
npm install

# Start the local development server
npm start
# Server runs at http://localhost:3000
```

Option 3: Running Automated Test Suites

```
# Run the complete test suite (149 tests)
node test_bare_task.mjs
node test_comprehensive.mjs
node test_stress.mjs
node test_checklist.mjs
```

Expected Output: **149 passed, 0 failed out of 149 tests (100% pass rate).**

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7. 🏁 Conclusion

OmniVoice AI proves that voice assistants can be fast, reliable, and multilingual without sacrificing deterministic control or privacy. By combining client-side hybrid NLU, proactive multi-turn context resolution, active alarm notifications, and full testing coverage, OmniVoice delivers a complete, production-ready hackathon project ready for judging.